

# TASK 3 — BODY SEGMENTATION METHOD TESTING

**Name:** Praneetha AS (aspraneetha29@gmail.com)

**Role:** AI/ML Intern

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## 1. Aim of the Study

The objective of this task was to study how Artificial Intelligence and Computer Vision systems separate the human body from the background using Body Segmentation techniques.

The task focused on understanding segmentation workflows, AI masking systems, and their applications in fashion technology, virtual try-on systems, healthcare, gaming, and real-time computer vision applications.

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## 2. Introduction to Body Segmentation

Body Segmentation is a computer vision process where Artificial Intelligence separates the human body from the surrounding background in images or videos.

AI segmentation systems identify:

- human body shape
- body boundaries
- clothing area
- foreground objects
- background regions

This technology is widely used in:

- virtual try-on systems
- AI fashion technology
- gaming applications
- augmented reality systems
- fitness tracking systems

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### **3. Working Principle of Body Segmentation**

AI body segmentation systems work in multiple stages.

#### **Step 1 — Image Input**

The camera captures the image or video frame.

The AI system receives:

- body posture
- clothing appearance
- body shape
- movement information

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#### **Step 2 — Human Detection**

Computer vision models first identify the presence of a human body.

AI algorithms locate:

- face
- shoulders
- torso
- hands
- legs

This helps isolate the person from other objects.

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#### **Step 3 — Segmentation Mask Generation**

The segmentation model generates a mask around the body.

The mask separates:

- human body
- clothes
- foreground
- background

The segmented area becomes easier for AI analysis.

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## Step 4 — Body Region Processing

The AI system processes:

- upper body
- lower body
- clothing regions
- body contours

This improves virtual fitting and body tracking accuracy.

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## Step 5 — Real-Time Tracking

The segmented body is tracked continuously in real time.

This enables:

- virtual clothing overlay
- AI fitness systems
- motion analysis
- body measurement estimation



### Segmentation Mask Output

The binary segmentation mask generated by the AI model.

#### Observation

- White region represents detected human body
- Black region represents removed background

#### Importance

This confirms that the AI correctly identified body regions.

### Background Removed Output

The final segmented image after removing the background.

#### Observation

- Human body remains visible
- Background is removed successfully

#### Importance

This demonstrates successful body isolation.



#### 4. Technologies Used in Body Segmentation

| Technology              | Purpose                 |
|-------------------------|-------------------------|
| Artificial Intelligence | Smart body analysis     |
| Computer Vision         | Human body detection    |
| Deep Learning           | Segmentation prediction |
| OpenCV                  | Image processing        |
| MediaPipe               | Pose tracking           |
| TensorFlow              | AI model training       |

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#### 5. Applications of Body Segmentation

Body segmentation technology is used in:

- AI fashion systems
  - virtual fitting rooms
  - gaming and animation
  - healthcare monitoring
  - augmented reality systems
  - fitness applications
  - motion tracking systems
  - AI surveillance systems
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#### 6. Advantages of Body Segmentation

The advantages include:

- accurate body isolation
  - better AI tracking
  - improved pose estimation
  - enhanced clothing overlay
  - real-time analysis
  - intelligent body mapping
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7. Challenges in Body Segmentation

| Challenge           | Description                                   |
|---------------------|-----------------------------------------------|
| Complex Backgrounds | Reduces segmentation accuracy                 |
| Poor Lighting       | Affects body visibility                       |
| Fast Movement       | Causes tracking instability                   |
| Loose Clothing      | Makes body shape detection difficult          |
| Occlusion           | Hidden body parts reduce segmentation quality |

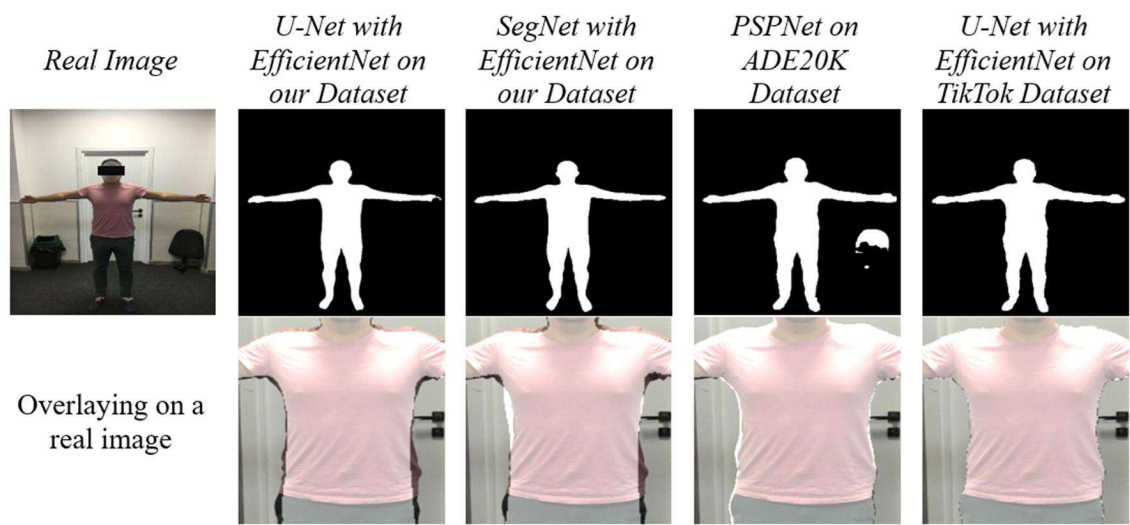
8. AI Models Used for Segmentation

Popular AI segmentation models include:

- MediaPipe Selfie Segmentation
- DeepLab
- Mask R-CNN
- OpenPose
- TensorFlow Segmentation Models

These systems are used for:

- body isolation
- AI masking
- virtual fitting applications



## **9. Learning Outcomes**

During this task, the following concepts were learned:

- body segmentation workflows
  - segmentation masking
  - AI body analysis
  - foreground extraction
  - computer vision systems
  - virtual try-on processing
  - real-time body tracking
  - segmentation model applications
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## **10. Conclusion**

Body Segmentation is an important technology in Artificial Intelligence and Computer Vision systems. It allows AI models to accurately separate the human body from the background for better analysis and tracking.

This task improved understanding of:

- segmentation systems
- AI masking workflows
- virtual fitting technology
- computer vision applications
- body tracking systems
- AI fashion technology